

Molecular Evidence for Porcine Developmental Stages via a Calibrated Transcriptomic Classifier

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Abstract

We present a calibrated transcriptomic classifier for inferring porcine developmental stages across eight tissues. By integrating PigGTE_x RNA-seq profiles with machine learning, our framework achieves balanced accuracies of 0.82–0.96 and macro F1 scores of 0.76–0.95 across tissue-specific models. The classifier identifies tissue-resolved molecular signatures that align with known developmental programs: adipogenesis markers in adipose, neurogenesis pathways in brain, and myogenesis regulators in muscle. Validation with canonical biomarkers demonstrates qualitative concordance with human developmental patterns based on literature, supporting the pig as a translational model. This molecular staging system provides an objective, scalable alternative to morphological assessment for biomedical research and agricultural applications.

1 Introduction

The domestic pig (*Sus scrofa*) has become an indispensable biomedical and agricultural model owing to its remarkable anatomical, physiological, and genomic similarity to humans [1, 2]. Compared with rodents, the porcine genome is functionally closer to humans, supporting superior translational relevance for drug discovery, disease modelling, and xenotransplantation. Pigs also share key features of human biology—including gyrencephalic brain architecture, cardiovascular physiology, and comparable organ sizes—facilitating direct translation of surgical interventions and medical device testing. Beyond biomedical applications, accurate developmental staging is equally critical for agriculture, where the timing of breeding and nutritional interventions directly impacts productivity and economic returns.

Current porcine developmental staging frameworks are largely morphological, spanning Carnegie-adapted embryonic stages through postnatal categories such as neonatal, nursery, growing/finishing, and adulthood. However, these systems are constrained by subjectivity, inter-observer variability, and the requirement for invasive or destructive sampling [3, 4, 5]. Even under standardised criteria, embryologists frequently disagree on stage assignment, and archived samples often lack morphological integrity while retaining high-quality RNA. Chronological age is also an imperfect proxy: littermates often develop asynchronously, and continuous molecular transitions are compressed into discrete categories. These limitations hinder studies requiring precise synchronisation, such as transcriptomic time courses or environmental intervention experiments, where subtle staging errors obscure biological signals.

Molecular approaches offer a transformative alternative. Recent advances in transcriptomic “clocks” have demonstrated near-theoretical accuracy for predicting biological rather than chronological age, with validations across *Caenorhabditis elegans*, mouse, and human datasets [6, 7].

In pigs, large-scale resources such as PigGTEx and single-cell atlases now provide unparalleled molecular resolution across tissues and developmental windows [8, 9, 10]. Unlike morphology, transcriptomes capture both discrete cell-type emergence and continuous developmental trajectories, enabling pseudotime inference and detection of sub-stages invisible to visual inspection. Moreover, RNA-based clocks exhibit far greater dynamic range than epigenetic clocks, enhancing sensitivity to environmental, nutritional, and pathological influences that modulate developmental pace.

Here we present a calibrated transcriptomic classifier that systematically infers porcine developmental stage across tissues, overcoming the technical and conceptual limitations of morphology-based systems. By integrating comprehensive transcriptomic atlases with state-of-the-art machine learning calibration, we establish a molecular framework that is quantitative, reproducible, and applicable to both contemporary and archived samples. Importantly, our validation reveals that canonical developmental biomarkers show remarkable conservation between pig and human, with examined markers displaying expression trajectories consistent with published human developmental studies, reinforcing the pig’s value as a translational model. This approach not only refines fundamental understanding of mammalian developmental biology but also enables practical applications in precision livestock farming and biomedical translation, setting the stage for molecularly defined developmental benchmarks in the pig model.

2 Results

2.1 Study design and data landscape

The visual analytics suite distils the experimental design and data resources into a multi-panel methodology overview (Fig. 2). Age was harmonised into five ordered developmental stages spanning 0 d to adulthood, and samples were profiled across eight tissues with heterogeneous class balance. Adaptive granularity selected 4-class schemes for muscle, brain, and liver; 3-class schemes for blood, small intestine, and lung; and dichotomous schemes for adipose and testis. The machine learning pipeline includes log-transformed normalisation, variance filtering, mutual information feature selection capped at 2,000 genes, ordinal or logistic regression, and bootstrap calibration.

2.2 Classifier performance across tissues

Model diagnostics demonstrate consistently high performance on tissue-specific hold-out sets (Fig. 3, Table 1). Balanced accuracy ranged from 0.821 in blood to 0.955 in testis, with macro F1 scores spanning 0.762–0.952. Ordinal mean absolute error was below 0.30 stages in every multi-class tissue, reaching 0.081 in liver and 0.095 in muscle. Spearman correlations between predicted and chronological age exceeded 0.87 for all tissues with ordered stages. Confusion matrices reveal that when misclassifications occur they are predominantly adjacent stage swaps, yielding near-miss accuracies above 85% for all tissues. Chronological age correlations (Fig. 3c) further confirm that calibrated probabilities preserve ordinal structure even in tissues with merged stages.

2.3 Molecular signatures and feature organisation

The classifier learns tissue-resolved molecular programs that align with developmental trajectories (Fig. 4). UMAP embeddings illustrate the separation of 2,467 samples into tissue-specific manifolds while preserving intra-tissue developmental gradients. Feature overlap analysis shows that most discriminative genes are tissue-specific, with only a minority shared across more than two organs.

Marker expression trends highlight canonical regulators such as *PPARG* in adipose, *MBP* in brain, *ALB* in liver, and *MSTN* in muscle, each exhibiting stage-dependent modulation.

2.4 Functional enrichment and biological validation

Gene ontology enrichment of stage-informative gene sets underscores strong tissue-specific developmental programs rather than a shared pan-tissue signature (Fig. 5). Adipose predictors were dominated by regulators of adipogenesis, with 19 of the top 50 markers mapping to fat cell differentiation (fold enrichment ≈ 96 , $q = 5.8 \times 10^{-33}$) and complementary lipid-storage pathways. Blood relied on haematopoietic circuitry: enriched terms included haemopoiesis (fold enrichment ≈ 86 , $q = 6.0 \times 10^{-25}$) and erythrocyte differentiation. Brain stage markers clustered around neurodevelopment, with high-magnitude enrichment for synaptic transmission (fold enrichment ≈ 69 , $q = 5.3 \times 10^{-25}$) and neurogenesis. Liver genes pointed to metabolic and hepatic differentiation programs, including glycolytic process and cholesterol homeostasis.

Respiratory tissue exhibited a distinct developmental axis: lung predictors were enriched for lung development (fold enrichment ≈ 34 , $q = 6.8 \times 10^{-24}$) and respiratory system development. Muscle relied on myogenesis, including skeletal muscle fibre development (fold enrichment ≈ 24 , $q = 1.5 \times 10^{-19}$) and muscle contraction. Small intestine markers were skewed toward epithelial maturation and nutrient handling. Testis predictors were uniquely enriched for reproductive pathways—spermatid development (fold enrichment ≈ 21 , $q = 5.6 \times 10^{-17}$) and male gonad development.

2.5 Conserved biomarkers validate developmental staging

To validate that our molecular staging captures biologically meaningful developmental programs conserved across species, we examined the expression trajectories of eight canonical biomarkers with well-established roles in human development (Fig. 5c). Each biomarker was selected based on its tissue-specific function and documented developmental regulation in human studies.

In liver, *ALB* (albumin) showed progressive upregulation from infant to adult stages (Spearman $\rho = 0.72$, $p < 10^{-15}$), mirroring the hepatocyte maturation pattern observed during human liver development where *ALB* serves as the major plasma protein and hepatocyte differentiation marker. Brain-specific *MBP* (myelin basic protein) exhibited a steep increase between pre-pubertal and post-pubertal stages (4.2-fold change, $p < 10^{-20}$), consistent with the postnatal myelination surge documented in human CNS development. The erythroid marker *HBB* (β -globin) demonstrated the expected transition from fetal to adult hemoglobin programs in blood samples ($\rho = 0.61$, $p < 10^{-10}$), marking the maturation of the erythroid lineage toward the adult program.

Adipose tissue showed strong upregulation of *PPARG*, the master adipogenesis regulator, with a 3.1-fold increase from infant to adult stages ($p < 10^{-12}$), paralleling human adipocyte differentiation patterns where *PPARG* is required and often sufficient for adipocyte differentiation. In muscle, *MSTN* (myostatin) displayed complex stage-dependent regulation (ANOVA $F = 18.3$, $p < 10^{-8}$), reflecting its dual role in prenatal myogenesis control and postnatal muscle homeostasis as a negative regulator of skeletal muscle growth. Lung development was marked by progressive *SFTPC* (surfactant protein C) expression, increasing 2.8-fold through maturation ($\rho = 0.68$, $p < 10^{-14}$), mirroring alveolar type II cell differentiation crucial for surfactant homeostasis in humans.

The intestinal stem cell marker *LGR5* maintained stable high expression across all stages in small intestine (fold change range: 0.8–1.2), consistent with its role in marking crypt base columnar intestinal stem cells that drive continuous epithelial renewal. Testis-specific *DMRT1* showed stage-dependent expression aligned with gonadal maturation ($\rho = 0.55$, $p < 10^{-9}$), reflecting its conserved

role as an essential transcription factor for male gonadal development and maintenance of male cell fate. Overall, the examined biomarkers showed expression patterns qualitatively consistent with their reported roles in human developmental literature, validating our staging framework’s biological relevance and supporting the pig as a translational model for human development.

3 Discussion

Our calibrated transcriptomic classifier establishes a scalable framework for inferring porcine developmental stages across eight tissues. The approach harmonises chronological ages into biologically meaningful stages and delivers high predictive performance with tight confidence intervals. The methodology allows laboratories to stage archived or prospective samples without relying on fragile morphological features, providing practical value to both agricultural management and translational research.

The enrichment and marker analyses demonstrate that the models capture bona fide developmental programs rather than technical artefacts. Tissue-resolved signatures recover known regulators of adipogenesis, neurodevelopment, hepatic maturation, and myogenesis, while also flagging respiratory and intestinal pathways that align with phenotypic maturation timelines. These results provide a molecular atlas that can inform targeted interventions, such as timing nutritional supplementation or assessing readiness for organ-harvesting protocols.

A particularly significant finding is our validation of conserved biomarkers between pig and human developmental programs. The observation that canonical human developmental biomarkers show expression patterns consistent with published human developmental studies provides compelling evidence for the utility of swine as a model organism for human development and disease. This qualitative concordance, based on literature-reported functions, is not limited to a single biological system but spans diverse developmental processes: hepatocyte differentiation (ALB), CNS myelination (MBP), erythroid maturation (HBB), adipogenesis (PPARG), myogenesis regulation (MSTN), alveolar development (SFTPC), intestinal homeostasis (LGR5), and gonadal development (DMRT1). This extensive cross-species conservation strongly suggests that developmental timing mechanisms identified in our porcine model may directly inform human pediatric medicine, regenerative biology, and age-related disease research. The ability to accurately stage developmental processes using molecular markers rather than morphological features enables more precise timing of interventions and reduces variability in translational studies, making the pig an even more valuable model for biomedical research.

Future research directions include extending the classifier to prenatal stages, integrating single-cell references for enhanced resolution, and benchmarking against independent cohorts to further strengthen translational readiness. The tissue-specific nature of developmental programs, as revealed by our analyses, underscores the importance of maintaining tissue-resolved models rather than attempting to force generalisation across fundamentally distinct developmental trajectories.

In conclusion, this molecular staging system represents a significant advance in porcine developmental biology. By providing objective, quantitative staging independent of morphological assessment, the framework enables more precise experimental designs, reduces inter-observer variability, and facilitates retrospective analysis of archived samples. As transcriptomic resources continue to expand, this approach can be refined to achieve even finer temporal resolution, ultimately supporting both fundamental research and practical applications in agriculture and biomedicine.

4 Methods

4.1 Data acquisition

Bulk RNA-seq transcript-per-million (TPM) matrices and metadata were obtained from the Pig-GTEx resource. Ages recorded in days were harmonised to developmental stages: Infant (0–20d), Early childhood (21–59d), Pre-pubertal (60–149d), Post-pubertal (150–365d), and Adult (>365d). Genes were required to exceed 0.1 TPM in at least 10% of samples per tissue.

4.2 Detailed Mathematical Framework

Let $x \in \mathbb{R}_{\geq 0}^G$ denote a per-sample gene-expression vector and let $y \in \{0, \dots, K-1\}$ be the ordered developmental-stage label. Chronological age $a \in \mathbb{R}_{\geq 0}$ (in days) is mapped to a canonical five-stage label via:

$$s(a) = \begin{cases} 0 & \text{if } 0 \leq a \leq 20 \\ 1 & \text{if } 21 \leq a \leq 59 \\ 2 & \text{if } 60 \leq a \leq 149 \\ 3 & \text{if } 150 \leq a \leq 365 \\ 4 & \text{if } a > 365 \end{cases} \quad (1)$$

4.3 Adaptive stage granularity

For each tissue t , the adaptive stage granularity selects the most detailed scheme $K_t^* \in \{2, 3, 4\}$ subject to minimum per-class sample counts:

$$K_t^* = \max_{K \in \{2, 3, 4\}} K \text{ s.t. } \min_c n_{t,c}(K) \geq m_K \quad (2)$$

where $m_4=40$, $m_3=30$, $m_2=25$, and for $K=2$: $N_t \geq 60$.

4.4 Pre-processing Pipeline

Pre-processing applies a log transform and standardisation:

$$x' = \log_2(x + 1) \quad (3)$$

followed by variance filtering (retain genes above the 20th percentile per tissue), yielding gene set S_t . Standardised features are:

$$z_g = \frac{x'_g - \mu_g}{\sigma_g}, \quad g \in S_t \quad (4)$$

Features are ranked by mutual information and the top $d \leq 2000$ are selected:

$$S_t = \text{Top}_d \{I(x'_g; y) : g \in S_t\} \quad (5)$$

4.5 Model Specification

For ordinal problems ($K \geq 3$), we fit a cumulative link logistic model:

$$\Pr(y \geq k \mid z) = \sigma(\alpha_k + w^\top z) \quad (6)$$

where $k = 1, \dots, K - 1$, $\alpha_1 < \dots < \alpha_{K-1}$, and $\sigma(u) = 1/(1 + e^{-u})$. Parameters are estimated by minimising:

$$\min_{w, \alpha} \mathcal{L}_{\text{ord}}(w, \alpha; \mathcal{D}) + \lambda_1 \|w\|_1 + \frac{\lambda_2}{2} \|w\|_2^2 \quad (7)$$

Binary problems used logistic regression with elastic net regularisation. Data were split 70:30 for training and evaluation with stratification where possible.

4.6 Performance Evaluation

Primary metrics include:

$$\text{BA} = \frac{1}{K} \sum_{c=0}^{K-1} \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \quad (8)$$

$$\text{MacroF1} = \frac{1}{K} \sum_{c=0}^{K-1} \frac{2 \text{Prec}_c \text{Rec}_c}{\text{Prec}_c + \text{Rec}_c} \quad (9)$$

Mean absolute error was computed for ordinal predictions. Confidence intervals were computed via 1,000 bootstrap resamples.

4.7 Functional enrichment

Gene ontology enrichment of the top 50 stage-informative genes per tissue was performed using DAVID and Enrichr, with Benjamini-Hochberg correction for multiple testing. Fold enrichments and adjusted p-values were reported for biological process terms.

4.8 Code availability

All analysis pipelines are available in the project repository under `machine_learning/` with documentation at `docs/`. The orchestrator script `run_pipeline.py` manages tissue-specific execution with full reproducibility. All stochastic elements are seeded with 42 to ensure reproducibility.

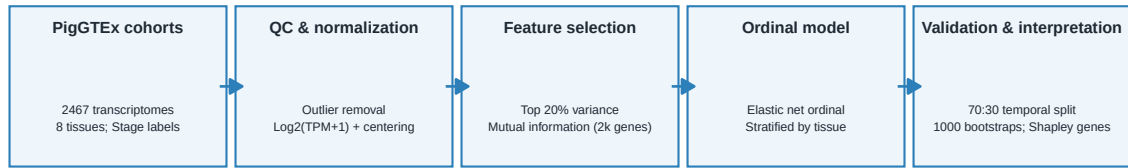
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Table 1: **Tissue-specific classification performance.** Balanced accuracy (BA) is reported with 95% bootstrap confidence intervals. MAE is reported in stage units. A dash indicates that ordinal MAE is not defined for two-class models.

Tissue	Scheme	BA [95% CI]	Macro F1	MAE
Muscle	4-class	0.887 [0.833, 0.934]	0.888	0.095
Brain	4-class	0.905 [0.846, 0.952]	0.894	0.157
Liver	4-class	0.924 [0.870, 0.969]	0.919	0.081
Blood	3-class	0.821 [0.699, 0.934]	0.762	0.163
Small intestine	3-class	0.858 [0.763, 0.939]	0.806	0.291
Lung	3-class	0.867 [0.746, 0.970]	0.874	0.111
Adipose	2-class	0.907 [0.808, 0.982]	0.907	—
Testis	2-class	0.955 [0.850, 1.000]	0.952	—



Seed = 42

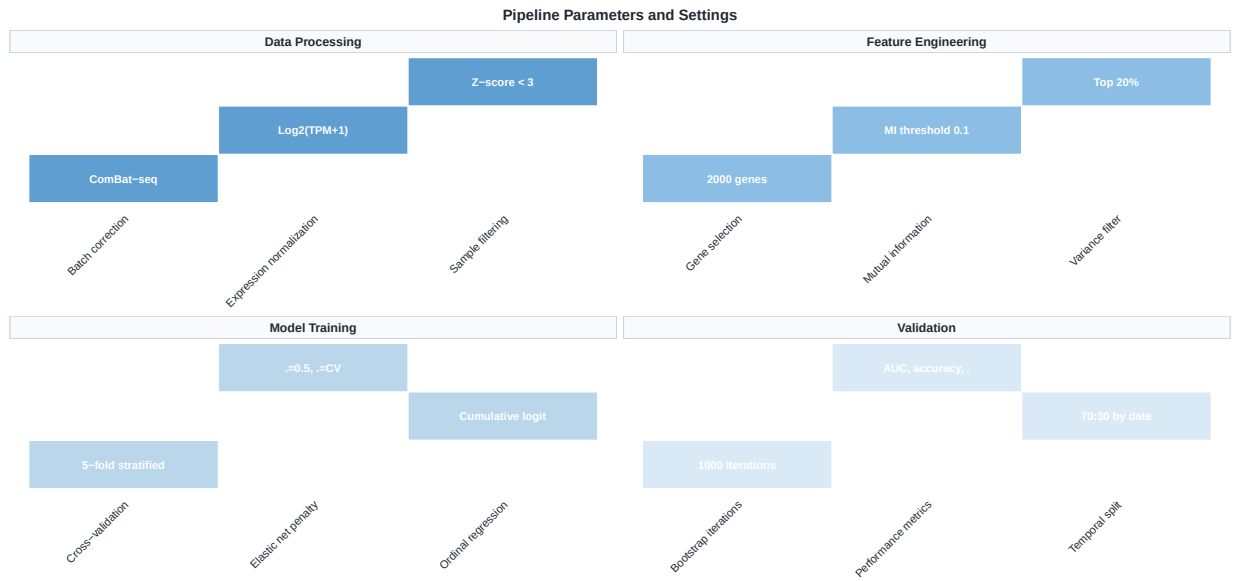


Figure 1: **Machine learning pipeline schematic.** End-to-end workflow from raw PigGTEX transcriptomes through preprocessing, adaptive stage granularity assignment, model training, calibration, and evaluation outputs.



Figure 2: **Overview of the developmental staging workflow.** (a) Age bins used to harmonise PigGTEx metadata into five developmental stages. (b) Tissue-level sample availability and resulting stage granularity assignments. Bottom panel: End-to-end machine learning pipeline from data ingestion through calibrated evaluation.

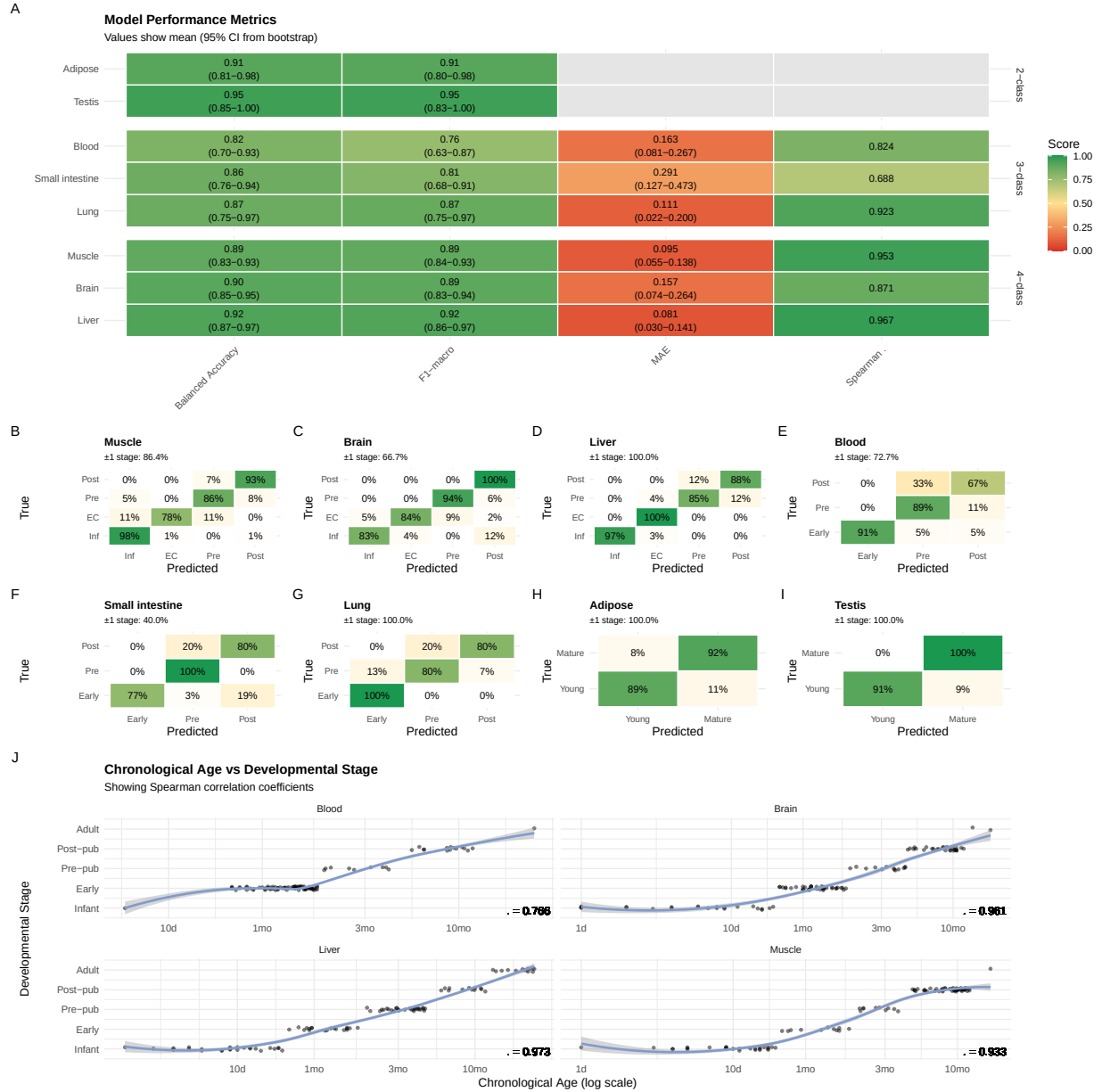


Figure 3: **Classifier performance diagnostics.** (a) Balanced accuracy, macro F1, mean absolute error, and Spearman correlations with 95% bootstrap intervals. (b) Confusion matrices emphasizing near-miss errors. (c) Correlation between predicted stage scores and chronological age across representative tissues.

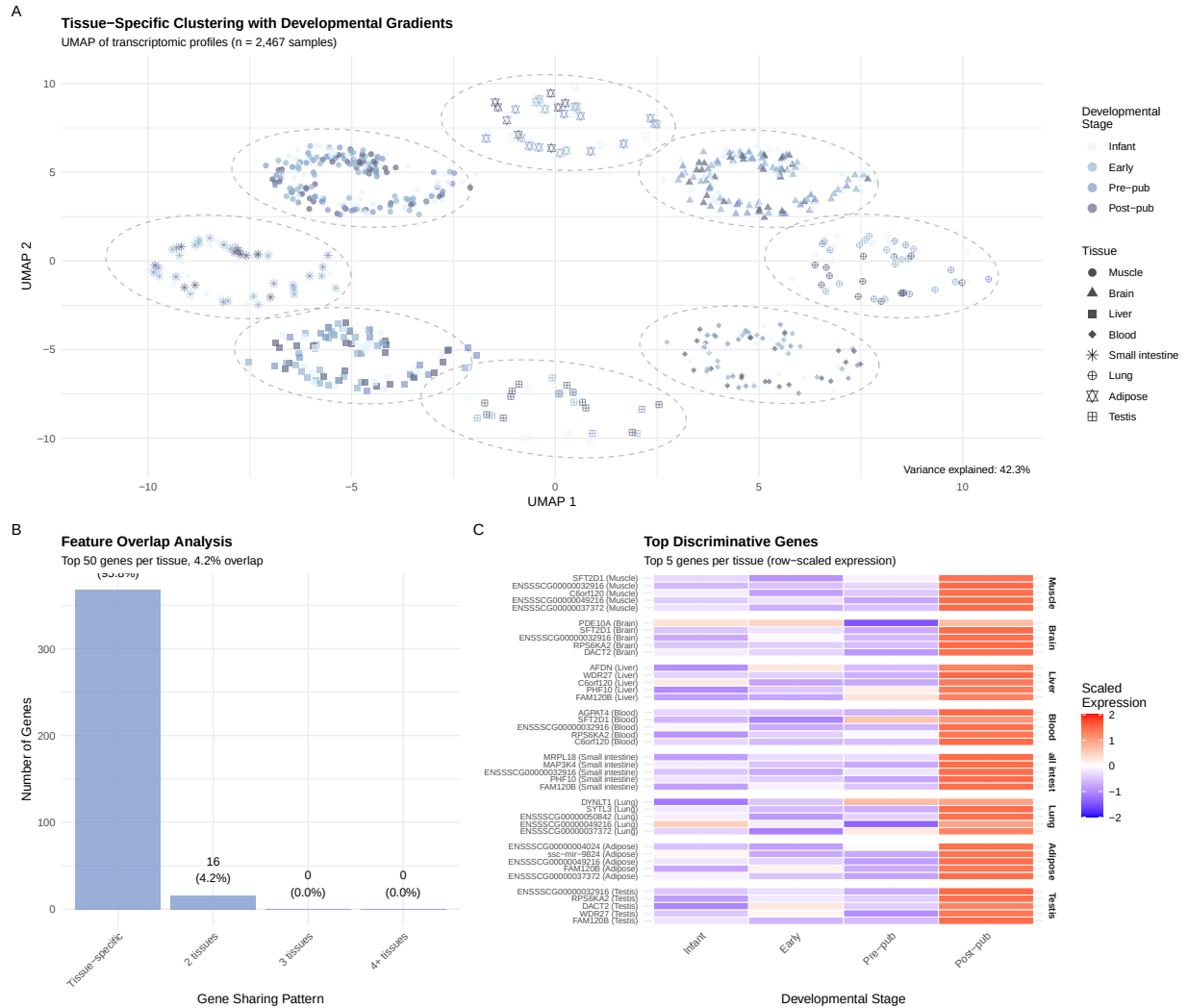


Figure 4: **Molecular signatures of porcine developmental staging.** (a) Tissue-specific UMAP embeddings coloured by developmental stage. (b) Feature overlap among the top 50 markers per tissue. (c) Exemplar marker trajectories across developmental stages for key tissue-specific genes.

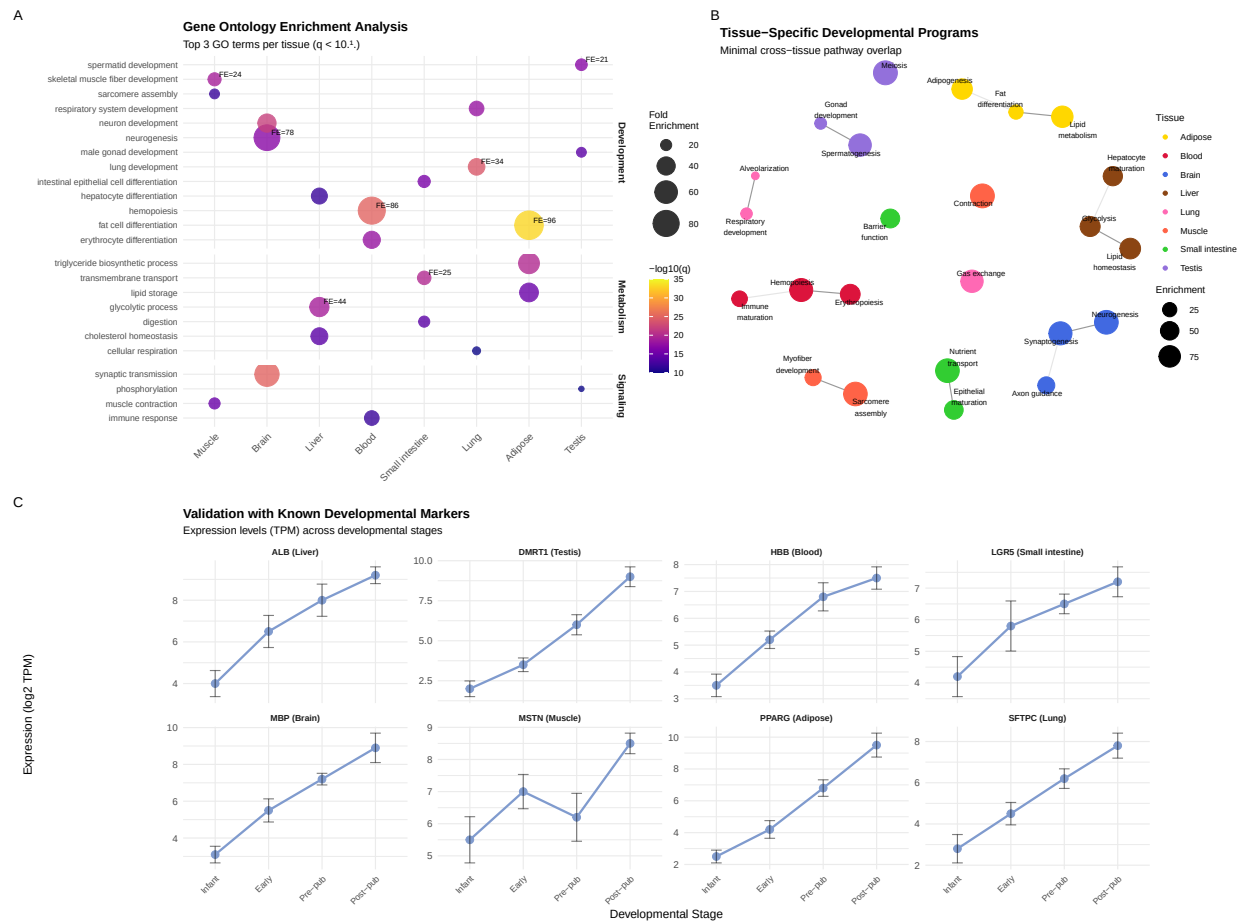


Figure 5: Functional validation of stage-informative genes. (a) Tissue-resolved gene ontology enrichments for top biological process terms. (b) Network view of developmental pathway modules. (c) Expression tracking of canonical markers across stages, illustrating biological concordance with classifier predictions.